

CATHAL O'CONNOR

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EDUCATION

THE GEORGE WASHINGTON UNIVERSITY, School of Engineering and Applied Science Washington, DC
Masters of Science, Computer Science Expected Graduation May 2027
Bachelor of Science, Computer Science May 2026
Minor in Economics
Technical GPA: 3.9

IMG ACADEMY Bradenton, FL
High School Diploma May 2021

EXPERIENCE/SKILLS

AI AND SOFTWARE INTERN

Ayata May 2026 - Aug 2026

- Built an ETL pipeline to process and transform operational midstream pipeline data into a PostgreSQL database, enabling network analysis and predictive modeling
- Collaborated with team members based in Canada and India

Ayata May 2025 - Aug 2025

- Created system monitoring tools to detect underperformance and network disruptions
- Built an AI simulator that enables operators to optimize oil production by running scenario-based simulations
- Collaborated with team members based in India and Texas

COMPOSITE OPERATING SYSTEM RESEARCH

George Washington University Jan 2026 - May 2026

- Integrated HPET (High Precision Event Timer) and I/O APIC support into the kernel for timer management and interrupt routing
- Developed low level kernel code for hardware initialization and interrupt controller configuration
- Worked alongside a Ph.D. candidate, faculty advisor, and undergraduate researcher to evaluate operating system design decisions and implementation strategies.

RTX AUTONOMOUS VEHICLE COMPETITION

George Washington University Interdisciplinary Sep 2025 - May 2026

- Developing software for autonomous drone and ground vehicle collaboration and communication
- Working with mechanical and electrical engineering students to integrate software and hardware
- Serving as Team Lead for the Computer Science division, overseeing agile development progress and task delegation

Coding Languages: C, Python, SQL, Java

ACTIVITIES

RUGBY REFEREEING

Potomac Rugby Referees Washington, DC
Referee October 2023 - Present

- Communicating clearly in high pressure situations while acquiring abilities to make quick decisions

RUGBY - GWU MENS RUGBY CLUB

Player Washington, DC
September 2023 - May 2026

- Compete in intercollegiate rugby matches while balancing athletic and academic commitments
- Lead team strategy and on field communication

LEAGUE OF LEGENDS / ESPORTS LEADERSHIP

GWU Esports League of Legends Organizer/Captain Washington, DC
September 2023 - May 2026

- Coordinated and managed all team operations, including scheduling practices, scrimms, and tryouts
- Created teams by grouping players with similar goals and expectations